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Research article

Techno-economic-environmental assessment of a hybrid renewable polygeneration system for an off-grid rural health clinic

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ABSTRACT

Rural health clinics often face severe challenges in ensuring reliable and sustainable energy supply due to the lack of grid access and limited fuel availability. This study presents the design and performance assessment of a fully self-sufficient hybrid renewable polygeneration system for an off-grid rural health clinic. The proposed configuration integrates building-mounted photovoltaic/thermal collectors, hydrogen energy storage, and a biomass furnace to meet the facility's electricity, space conditioning, and domestic hot water demands. The system is dynamically simulated in TRNSYS, and a comprehensive techno-economic-environmental analysis is conducted to evaluate its performance. Results show that the system meets the clinic's annual 37 MWh energy demand with a negligible loss of load probability of 0.37%, maintaining indoor thermal comfort year-round. The total life-cycle cost is approximately $141,670 and the levelized cost of energy equal to $0.3/kWh. Compared to a diesel-based alternative the system achieves a twelvefold reduction in life-cycle carbon emissions. Additionally, it produces 3755 m^3 of oxygen annually via water electrolysis, providing a potentially valuable medical resource for the facility. Overall, the findings demonstrate that a fully renewable, high-reliability energy system can be economically and environmentally viable for off-grid healthcare applications in arid climates, offering a scalable model for broader deployment.

1. Introduction

Access to reliable and sustainable energy is a central pillar of the Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy). Despite ongoing grid expansion efforts, extending conventional electricity networks to remote rural areas remains economically prohibitive and environmentally burdensome. As a result, more than 700 million people, primarily in sub-Saharan Africa and developing Asia, still lack access to electricity, constraining socio-economic development and access to essential services such as healthcare [1,2].

Reliable energy supply is indispensable for the operation of rural health clinics (RHCs), supporting critical services including medical equipment, vaccine refrigeration, lighting, and heating, ventilation, and air-conditioning (HVAC) systems. In many off-grid settings, RHCs depend on diesel generators that are costly, polluting, and unreliable, or operate with limited or intermittent power availability [3,4]. These constraints compromise patient care, restrict operational capacity, and negatively affect health outcomes in already vulnerable communities.

Renewable energy systems (RES) offer a decentralized and sustainable alternative for electrifying remote healthcare facilities by utilizing locally available resources [5,6]. Technologies such as solar photovoltaic (PV) systems [7,8], wind energy [9,10], and bioenergy derived from organic materials [11] have been widely explored for rural electrification. More recently, photovoltaic–thermal (PVT) and building-integrated photovoltaic–thermal (BIPVT) systems have gained attention due to their ability to simultaneously generate electricity and useful thermal energy while reducing land-use requirements [12,13].

To address the intermittency of individual renewable resources, hybrid renewable energy systems (HRESs) that integrate multiple generation technologies, often coupled with energy storage, have been increasingly investigated for off-grid applications. By exploiting the complementary nature of renewable resources, HRESs can enhance supply reliability, reduce environmental impacts, and improve cost-effectiveness compared with single-source systems [14–16]. Consequently, they represent a promising solution for powering remote RHCs.

Several studies have assessed HRES configurations for healthcare facilities in diverse geographic contexts. PV/wind-based systems,

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sometimes combined with diesel generators or battery storage, have been shown to reduce levelized cost of energy (LCOE) and carbon emissions compared with diesel-only systems [17–19]. Other works have explored hybrid configurations incorporating biogas [11] or batteries to improve reliability and reduce operating costs. However, the majority of these studies focus primarily on meeting electrical loads, with thermal energy demands either neglected or met indirectly through electrical devices.

Energy storage plays a critical role in enabling reliable HRES operation. While electrochemical batteries are commonly used for short-term storage [20], their suitability for long-duration or seasonal storage is limited. Hydrogen energy storage (HES) has emerged as a promising alternative, offering high energy density, long-duration storage capability, and zero direct emissions when produced from renewable electricity [21–24]. In healthcare applications, HES can enhance system resilience during extended periods of low renewable generation. Although battery storage generally offers substantially higher round-trip efficiency, this advantage does not necessarily mean greater reliability in autonomous systems. Previous research has shown that hydrogen storage can become more advantageous under high renewable penetration and strong seasonal energy imbalance, owing to its long-duration storage capability [25]. A recent comparative TRNSYS study also showed that, at the same LCOE, an HES-based configuration achieved zero Loss of Load Probability (LOLP) and considerably lower renewable energy curtailment than a battery-based configuration, despite the higher round-trip efficiency of the latter. This difference was attributed to the lower charging capability of the battery system due to interruptions during charging and discharging, which resulted in greater unused surplus electricity [26]. In addition, water electrolysis provides oxygen as a by-product, representing a potentially useful supplemental resource for healthcare facilities.

Several studies have investigated renewable and hydrogen-based polygeneration systems in recent years. For example, Saleem et al. [27] used TRNSYS to investigate a solar-hydrogen cogeneration system comprising a PV array, electrolyzer, fuel cell, battery, and hydrogen storage unit. The system achieved efficiencies of 7.8% under the colder, lower-solar-insolation conditions of Fargo and 11.8% under the warmer climate of Lahore, highlighting the influence of climatic conditions on system performance. In a subsequent solar-assisted grid-tied polygeneration system, a 27-kWp PV configuration produced 12,696 m^3 of hydrogen, 6348 m^3 of oxygen and 11.56 GJ of thermal energy annually, while exporting 16,848 kWh of electricity to the grid [28]. More recently, an optimized renewable-driven polygeneration system integrating solar thermal and PV technologies, wind energy, geothermal energy, and thermal, battery, and pressurized hydrogen storage was investigated using TRNSYS coupled with GenOpt. The optimized configuration achieved a solar fraction of 0.78, generated 8020 MJ of useful solar heat annually, and produced 12,696 m^3 of hydrogen per year [29]. Another TRNSYS-based polygeneration system combined 1 MW of solar PV, 1 MW of wind generation, and 1.5 MW of auxiliary diesel generation to meet varying electrical loads. The renewable sources supplied 77% of the total energy demand, while the remaining 23% was provided by diesel, and surplus renewable electricity was converted into approximately 5000–7000 kg of hydrogen for storage and later use [30]. From a broader assessment perspective, a multi-heat-recovery polygeneration system based on a gas-turbine Brayton cycle, Kalina cycle, PEM electrolyzer, and multi-effect desalination unit was evaluated using energy, exergy, economic, and environmental indicators. The system simultaneously produced electricity, hot and chilled water, hydrogen, and freshwater, achieving energy and exergy efficiencies of 74.98 and 32.15%, respectively, a fuel-saving potential of 28.99%, and a cost of energy of $0.38/kWh [31]. These studies demonstrate the potential of integrated polygeneration systems for simultaneously producing multiple energy products; however, their applications differ from the specific requirements of off-grid healthcare facilities, where reliable simultaneous supply of electricity, space conditioning, and DHW is essential.

As summarized in Table 1, existing studies that incorporate hydrogen storage in off-grid healthcare systems remain limited in number and scope, with many relying on diesel backup or grid connectivity.

As shown in Table 1, existing HRES-HES studies for healthcare facilities primarily focus on electricity supply, with limited consideration of thermal energy demands such as space conditioning and domestic hot water. Moreover, most configurations either rely on conventional PV systems without thermal recovery or remain partially dependent on diesel generators or grid support, particularly to ensure reliability during extended periods of low renewable generation.

Overall, several key research gaps remain. First, the lack of integrated treatment of electrical and thermal loads limits the ability of existing systems to realistically represent the full operational requirements of healthcare facilities. Second, although PVT and BIPVT systems offer clear advantages over conventional PV systems, their integration into HRES designs for rural healthcare applications has been largely overlooked. Similarly, the potential of biomass as a locally available and low-cost thermal energy source in rural settings remains underexplored. Third, research focusing on RHCs in hot and arid climates, where cooling demand is high but solar resources are abundant, is scarce.

This study addresses these gaps by proposing a fully renewable and self-sufficient HRES for an off-grid rural health clinic located in the hot and dry climate of Yazd, Iran. The proposed system integrates BIPVT for simultaneous electricity and heat generation, biomass for auxiliary thermal supply, and hydrogen-based energy storage to enhance system reliability and energy management. The system is comprehensively evaluated in terms of technical performance, economic feasibility, and environmental impact, with the objective of providing a sustainable energy solution tailored to underserved rural healthcare facilities.

Table 1
Overview of recent studies on HRES-HES for off-grid healthcare facilities.

Ref.	Location/Climate	Facility	HRES	Daily Load	Load Types	LCOE ($/kWh)	CO_2 Reduction
[32]	Eastern Cape, South Africa/ Subtropical	Rural health clinic (10-bed facility)	PV/wind	18.67 kWh	Electrical (medical equipment, lighting, and appliances)	0.234	Near 100% vs. Diesel
[33]	Konya, Turkey/Temperate continental	Animal hospital (veterinary clinic)	PV/wind	150–500 kWh (varies by building)	Electrical (HVAC, appliances, medical/lighting)	0.321	Up to 80% vs. Diesel
[34]	Istanbul, Turkey/ Mediterranean	Nursing home	PV/wind	110 kWh	Electrical (lighting, televisions, refrigerators, appliances, computers, electronic equipment)	1.306	N/A
[35]	Elands Bay, South Africa/ Mediterranean	Rural healthcare facility	PV	44.82 kWh	Electrical	1.49	100% vs. Diesel
[36]	Four climate zones (highland, lowland, arid, and coastal) in Pakistan	Hospital	PV	603.45–1278.4 kW	Electrical	0.3159	99% vs. Diesel

2. Materials and methods

This section begins with a detailed description of the RHC and the energy system designed to meet its energy requirements. Subsequently, the modelling approach implemented in TRNSYS is outlined, and performance assessment is explained. Fig. 1 presents the overall process flow and analysis framework adopted in this study. The methodology starts with the definition of the case study and assessment of the RHC's energy demands, followed by the HRES configuration and dynamic simulation in TRNSYS. The simulated performance is subsequently evaluated from technical, economic, and environmental perspectives.

2.1. Case study description

The remote off-grid RHC is located in Yazd, Iran. The location has an average daily global solar irradiation of 5 kWh/m^2 [37] and temperature variations between -5°C and 44°C throughout the year. The two-storey building is equipped with PVT modules on its south-facing wall and inclined roof, forming a BIPVT system. It has a ground floor area of 100 m^2, and its roof is inclined 20° towards the south.

The healthcare facility is designed to provide comprehensive patient care, with a focus on both clinical and diagnostic services. The centre is equipped for extensive laboratory analysis, allowing for pathological and haematological testing. It also includes dedicated spaces for physician consultations and patient recovery, with beds equipped with oxygen support to ensure a high standard of patient management and stabilisation. All electrical appliances and their operating schedules are presented in the Appendix of the paper (Table A1). The total daily electrical energy consumption of the included equipment is 32 kWh.

In addition to the energy required for the appliances listed in Table A1, space heating and cooling represent a considerable portion of the RHC's overall energy demand. Fig. 2 presents the monthly variations in heating and cooling loads necessary to maintain indoor thermal comfort (20 °C during cold months and 24 °C during warm months). Because the centre is located in an arid region with high solar radiation, its cooling demand far exceeds its heating requirement. The annual cooling demand is 9.43 MWh, compared to 3.56 MWh for heating. The peak cooling load of 2.4 MWh occurs in July, while the highest heating load of 1.2 MWh is recorded in January.

The thermal energy required for the provision of DHW at 50 °C, is also included in the overall energy demand assessment. Considering an average water flow rate of 50 L/h during peak hours and 20 L/h during off-peak hours, the total annual thermal energy demand for DHW is estimated at 11.85 MWh.

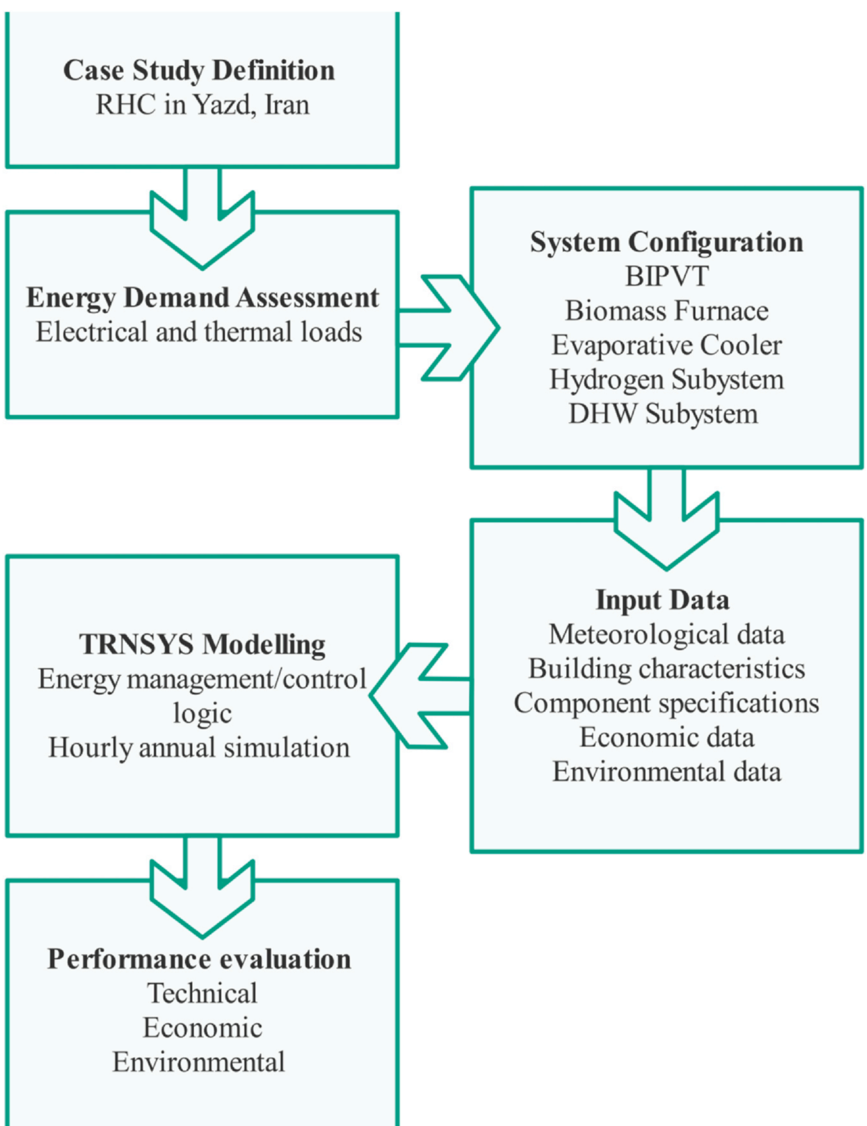

Fig. 1. Process flow and analysis framework of the study.

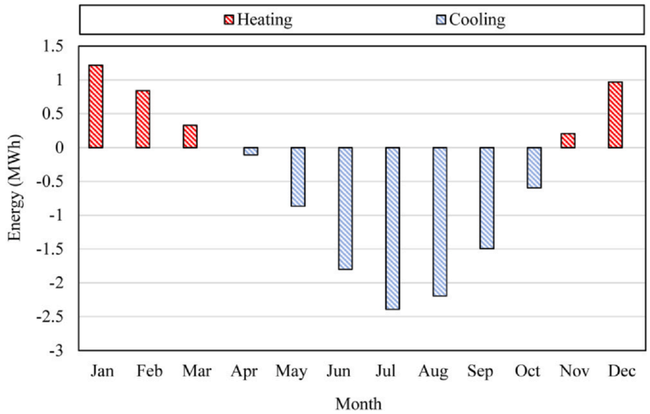

Fig. 2. Monthly space heating and cooling loads.

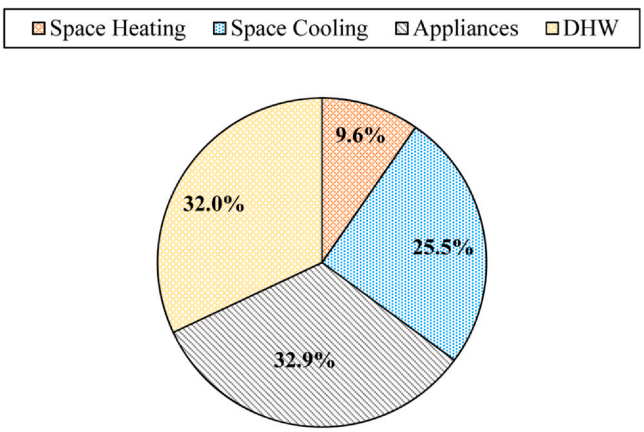

Fig. 3. Breakdown of the overall energy demand.

The distribution of energy requirements for the RHC under study is presented in Fig. 3. The power demand in this RHC ranges from 0.7 kW to 9.4 kW, depending on the time in the day and year. Daily energy demand also ranges from 60 kWh/day (during April) to 152 kWh/day (during July), and the total annual energy demand is 37 MWh. As shown, medical appliances account for the largest share of total energy consumption, followed by DHW, while space heating represents the smallest share.

2.2. Energy system description

The energy system is designed to fully meet the RHC's energy demands, including electricity, space heating and cooling, and DHW, with high reliability. The proposed system consists of PVT modules, a biomass-fired hot air furnace, an evaporative cooler, and an HES unit, as shown in Fig. 4. In addition to the primary energy functions, the HES unit also produces pure oxygen used for medical applications within the healthcare facility. The biomass furnace is fuelled by dried sheep manure supplied by local farmers, thereby generating additional income for the surrounding community. The technical specifications of the main system components are given in the Appendix of the paper (Table A2).

The facility's PVT modules contribute to both thermal and electrical loads. Outdoor air is drawn through a narrow channel behind the modules, where it is heated during passage. If the outlet air temperature is sufficiently high, it is used directly for space heating. When the heated air cannot fully satisfy the heating load, the biomass furnace is activated to cover the shortfall. During periods of low solar radiation, extremely cold conditions, or in warm seasons, the outlet air from the PVT modules is not used for heating but for the cooling of the modules. In warm seasons, when cooling is required, the evaporative cooler is activated to maintain indoor thermal comfort.

The operational logic of the space heating and cooling section is shown in Fig. 5a. Continuous monitoring of indoor and outdoor temperatures is crucial, as these measurements determine the control signals for switching components on and off to ensure reliable operation.

For electrical and DHW loads, the electrical output of the PVT modules is utilised. A controller continuously measures the generated power, managing its distribution among the system components. Any surplus electricity is directed to an alkaline electrolyzer (AEL) to produce hydrogen, which is then stored at 200 bar. When state of charge (SOC) of the hydrogen tank surpasses 90%, the electrolyzer is switched off. When the electrical generation is not enough to cover the demand, a proton exchange membrane fuel cell (PEMFC) is used to make up for the deficit. The fuel cell is shut down when the SOC of the hydrogen tank falls below 10%. In addition to hydrogen, the electrolyzer produces oxygen gas that is then compressed and stored in special cylinders for medical use.

The waste heat from the fuel cell is recovered through a cooling water circuit and transferred to the DHW tank. To guarantee a reliable DHW supply, an electric heating element is also installed in the tank to meet any remaining demand. The operational logic of the electricity and DHW supply section of the energy system is shown in Fig. 5b.

It is worth noting that the physical locations of the components shown in Fig. 4 are schematic and do not represent their actual siting within the healthcare facility. In practical implementation, compressed hydrogen storage at this pressure requires appropriate safety measures, including hydrogen leak detection and emergency shutdown systems, pressure-relief devices, separation from ignition sources and combustible materials, protection against unauthorized access and physical damage, and regular inspection and maintenance [38].

2.3. Modelling

The proposed energy system is modelled in TRNSYS 18, and its hourly performance over the entire year is simulated. The mathematical formulations and governing equations of the standard TRNSYS component models are provided in the TRNSYS 18 Mathematical References [39,40]. All operational logics presented in Fig. 5 are defined in TRNSYS (Fig. 6), and the components used in the simulation are listed in Table 2. To integrate the PVT panels into the building structure and form a BIPVT system, the back-sheet properties of the PVT panels are matched to the exterior characteristics of the building's outer wall.

Meteorological data for the period 2000–2019 for the Yazd weather station, Iran (31.9° N, 54.4° E, 1230 m a.s.l) were obtained from the Meteonorm 8 database.

To assess the reliability of the TRNSYS component models, the BIPVT, AEL, and PEMFC sub-models were compared with independent experimental data from the literature under comparable operating conditions. The validation conditions and corresponding deviations are summarized in Table 3. The observed deviations are mainly due to differences between the simplified or semi-empirical formulations used in the TRNSYS component models and the detailed characteristics of the experimental systems, including membrane and electrode properties, heat-transfer characteristics, and other test-specific parameters that are not explicitly represented in the models. Despite these differences, there is reasonable agreement with the experimental data.

2.4. Performance evaluation

This study evaluates the proposed system based on three key categories: technical, economic, and environmental performance. The specific indicators within each category are explained in the following subsections.

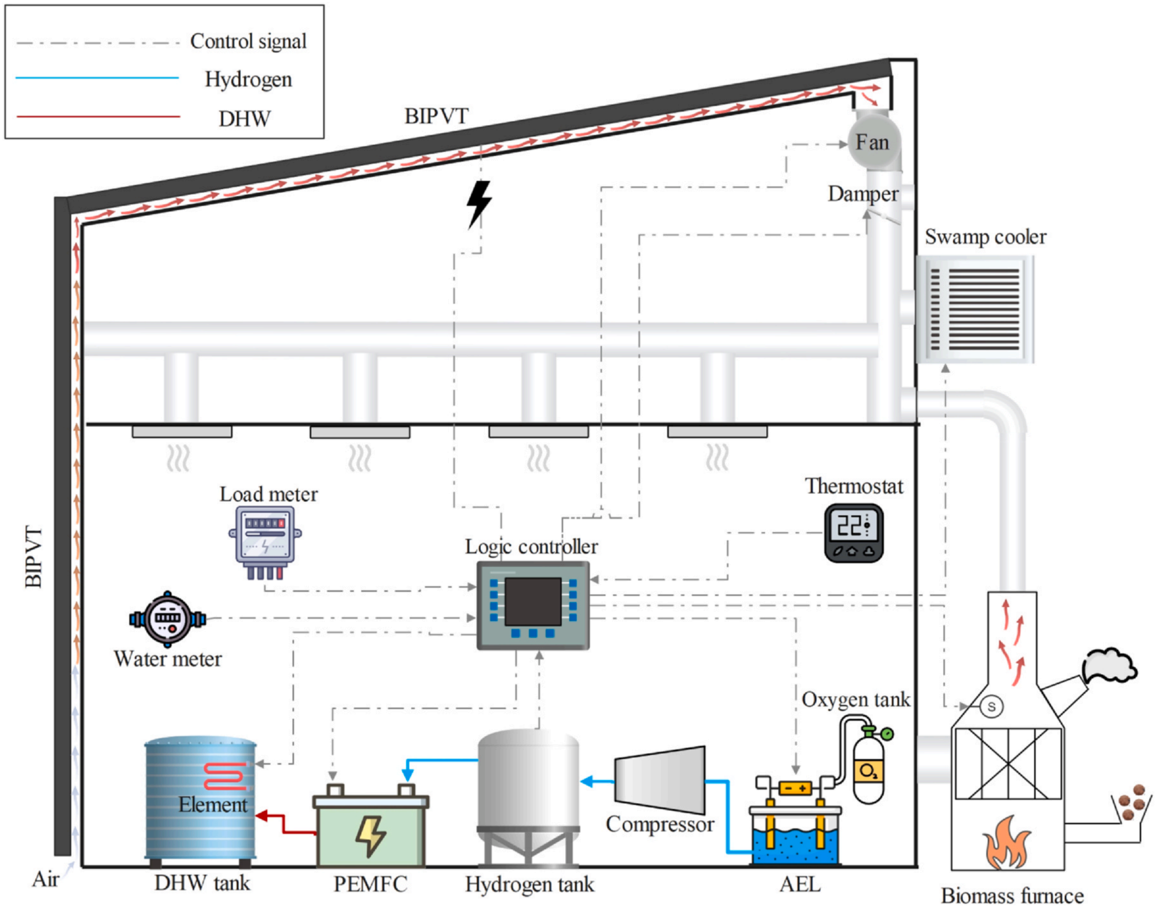

Fig. 4. Schematic representation of the proposed energy system.

2.4.1. Evaluation of the technical performance

Loss of Load Probability (LOLP) is a key technical indicator for assessing the reliability of an energy system. It quantifies the likelihood that the system will fail to meet the energy demand during a specific period. A lower LOLP value indicates higher system reliability, meaning the power supply is more robust and there is lower probability of not meeting the demand. The annual LOLP is defined as the ratio of the total energy deficit to the total annual energy demand [44]:

$$LOLP = \sum_{t=1}^{8760} ES(t) / \sum_{t=1}^{8760} LD(t) \quad (1)$$

where, ES (t) represents the energy shortage in a given hour t, and LD (t) is the corresponding load in that hour, both measured in kWh. In this study, LOLP is calculated for the overall energy demand of the RHC, including electricity (also supplying space cooling and partially DHW), space heating, and DHW. At each time step, the unmet demand of each load category is determined separately and the corresponding shortages are summed to obtain the total energy shortage. Similarly, the individual load demands are summed to obtain the total energy demand.

2.4.2. Evaluation of the economic performance

LCOE is widely used as a standardised economic indicator to compare the cost-effectiveness of different energy generation technologies on an equal basis. It represents the average cost of producing one kWh over the entire operational lifetime of a system. It is calculated by dividing the total annualised cost of the system (including the capital expenditure, the operation and maintenance costs, fuel expenses, and replacement costs) by the total energy supplied to the user in the same period [45]. The definition of LCOE is shown in Eq. (2).

$$LCOE = \left(\sum_{t=0}^{20} \frac{C_t + O_t + F_t}{(1+d)^t} \right) / \left(\sum_{t=0}^{20} \frac{E_t}{(1+d)^t} \right) \quad (2)$$

where, C_t is the total capital expenditure, O_t the annual operation and maintenance costs, and F_t the annual fuel costs. The fuel costs in this study refer to the cost of the biomass. E_t is the total annual energy output, while d is the discount rate.

Alongside LCOE, the Total Life-Cycle Cost (TLCC) is also used to evaluate the full costs of the project over its entire lifespan, from pre-development to decommissioning. All costs are discounted to a base year [46]. TLCC is estimated using Eq. (3).

$$TLCC = \left(\sum_{t=0}^{20} \frac{C_t + O_t + F_t}{(1+d)^t} \right) \quad (3)$$

Table 4 presents the cost functions of all system components. The system is evaluated over a 20-year lifespan. The fuel cell and electrolyzer are replaced after 40,000 and 90,000 hours of operation, respectively.

2.4.3. Evaluation of the environmental performance

In this study, the Carbon Footprint of Energy (CFOE) is used to evaluate the environmental performance of the proposed system, because it provides a direct measure of its contribution to climate change. CFOE represents the total carbon dioxide emissions associated with the system throughout its entire life cycle. It is expressed as the amount of greenhouse gases emitted per kWh of energy produced, covering all stages from manufacturing and installation to operation and decommissioning. This comprehensive metric enables a clear comparison of the system's environmental impact relative to alternative energy solutions [44].

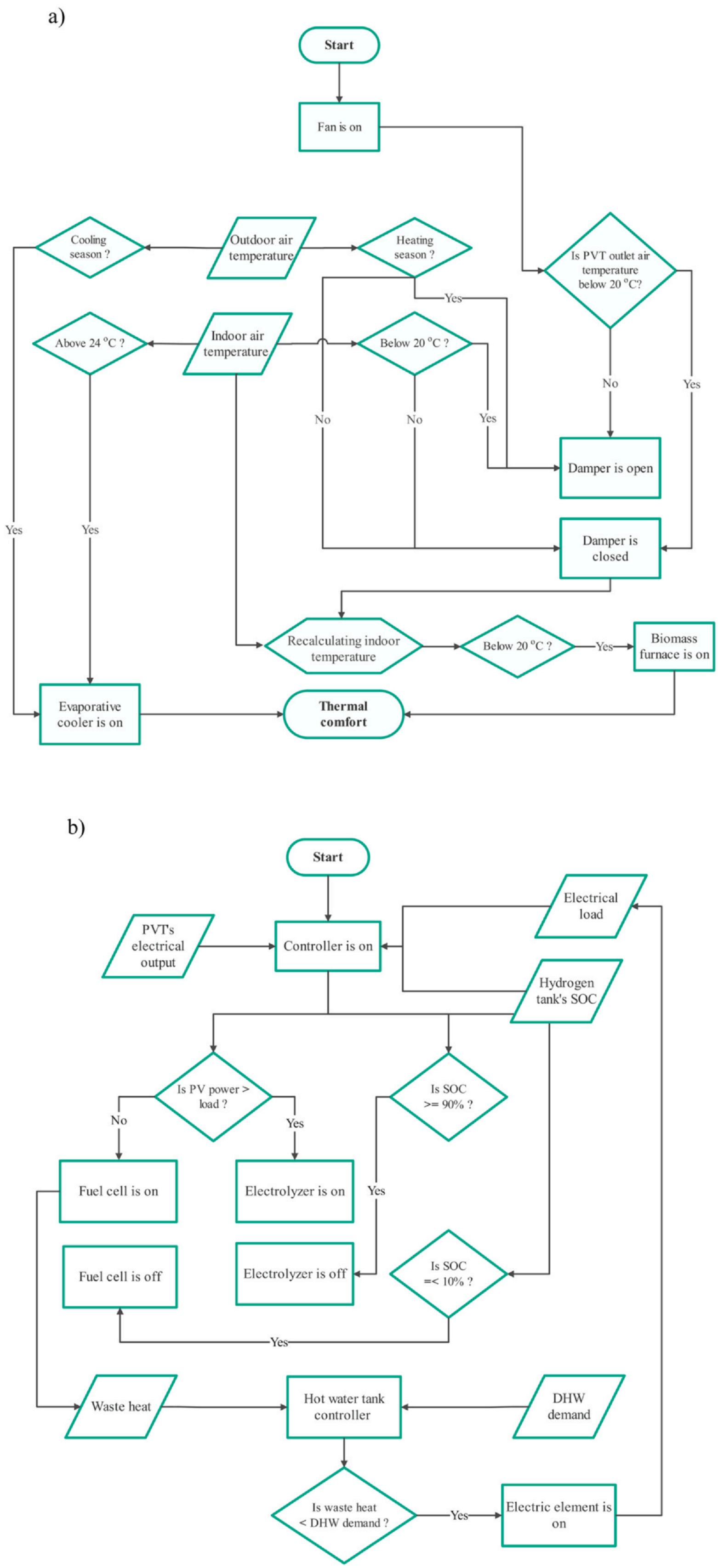

Fig. 5. Flowchart of a) the space heating and cooling section and b) the electricity and DHW section.

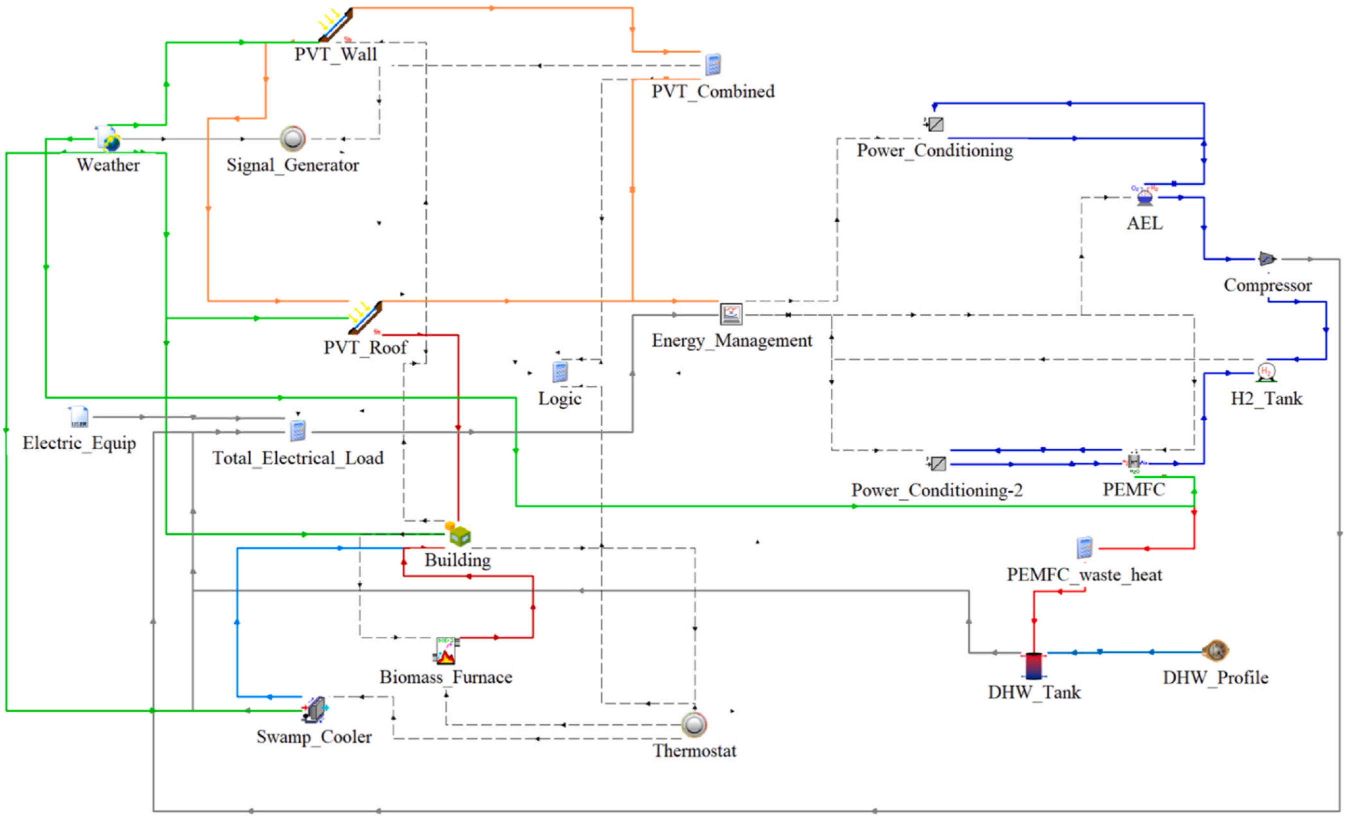

Fig. 6. TRNSYS simulation of the proposed energy system.

Table 2
TRNSYS components used in the simulation of the system.

Component	TRNSYS Type
BIPVT	567–5
AEL	160a
Biomass furnace	967
Building	56
Converter	175a
DHW tank	4a
Energy management controller	105a
Hydrogen compressor	167
Hydrogen tank	164b
PEMFC	170a
Evaporative cooler	506d
Thermostat	166
Water meter	14b
Weather	15–6

To assess how environmentally friendly the proposed energy system is, it is compared with a reference scenario, where the RHC relies on a conventional diesel generator for electricity supply, and a diesel boiler

Table 4
Cost functions of system components [47,48].

Component	Investment Cost (C)	Operation Cost (O)	Unit
PVT	$66.7 \times A_{PVT}$	1%C	$\$/m^2$
Alkaline electrolyzer	$1200 \times P_{AEL}$	2%C	$\$/kW$
PEM fuel cell	$2000 \times P_{PEMFC}$	2%C	$\$/kW$
Hydrogen tank	$300 \times m_{H_2}$	1%C	$\$/kg$
Hydrogen Compressor	$91562 \times 0.89\left(\frac{P_{comp}}{455}\right)^{0.67}$	1%C	$\$/kW$
Hydrogen piping	10000	-	$
Biomass furnace	$9000 + Q_{sh}$, f/4 where Q_{sh}, f is the heating energy supplied by the furnace (kWh)	1%C	$\$/kWh$
Evaporative cooler	1500	1%C	$
Sheep manure	$0.1 \times m$	-	$\$/kg$
DHW tank	2000	1%C	$
Miscellaneous	10% C_{total}	-	$

Table 3
Experimental validation of the main TRNSYS component sub-models.

Component	Validation conditions	Validation metric	Experiment	TRNSYS	Relative deviation	Ref
BIPVT-thermal	Total radiation = 939 W/m^2; Airflow = 85 kg/h; Ambient temperature = 26°C	Thermal efficiency	18.0%	18.8%	4.4%	[41]
BIPVT-electrical	Total radiation = 821 W/m^2; Airflow = 349 kg/h;	Electrical power density	115.7 W/m^2	106.4 W/m^2	8.0%	[41]
AEL	Operating temperature = 65°C; Pressure = 7 bar; Current density = 1974 A/m^2;	Cell voltage	1.85 V	1.78 V	3.8%	[42]
PEMFC	Operating temperature = 60°C; hydrogen inlet pressure = 1.2 bar; oxygen inlet pressure = 1 bar; hydrogen stoichiometric ratio = 1.4; oxygen stoichiometric ratio = 1.8	Cell voltage	0.433 V	0.375 V	13.4%	[43]

Table 5
Life-cycle carbon footprints of energy-producing components.

Component	CF Value (gCO_{2eq}/kWh)	Ref
PVT	47.156	[49]
PEMFC	37	[50]
Biomass furnace	200	[51]
Diesel generator	763	[52]
Diesel boiler	296	[53]

for space heating and DHW. The CFOE is calculated using Eqs. (4) and (5).

$$CFOE = \frac{CF_{PVT} \times E_{PVT} + CF_{PEMFC} \times E_{PEMFC} + CF_{Bio} \times E_{Bio}}{E} \tag{4}$$

$$CFOE_{ref} = \frac{CF_{DG} \times E_{DG} + CF_{DB} \times E_{DB}}{E} \tag{5}$$

where, CF denotes the life-cycle carbon footprint of each energy-producing component (Table 5), and E represents the total energy production amount in a specific period, such as a year.

3. Results and discussion

The performance of the proposed energy system is analysed in this section, covering technical, economic, and environmental aspects.

3.1. Technical assessment

Fig. 7 presents the breakdown of the electricity consumption in the RHC. Besides the appliances discussed in Section 2.1, the energy system must supply electricity for the DHW tank auxiliary heater, the hydrogen compressor in the HES unit, and the evaporative cooler. The electricity requirements for these components are obtained from the TRNSYS simulation. The total annual electricity consumption is 24.89 MWh.

As shown in Fig. 7, the electricity consumption peaks in July, primarily due to the increased cooling load and, consequently, the higher power usage of the evaporative cooler, along with elevated electricity consumption of the hydrogen compressor. Conversely, the electricity consumption of the DHW auxiliary heater is at its lowest in July and August, due to the extended operation of the fuel cell during this period. Apart from this electricity consumption, the remaining part of the overall energy demand is covered by the thermal output of the PVT system, the biomass furnace, and the waste heat of the fuel cell.

The proposed energy system delivers energy with a very low annual LOLP, equal to 0.37%. The space heating and cooling section successfully maintains thermal comfort throughout the year, as shown in Fig. 8. The hourly indoor temperature stays above 20°C in cold weather and below 24°C at all other times.

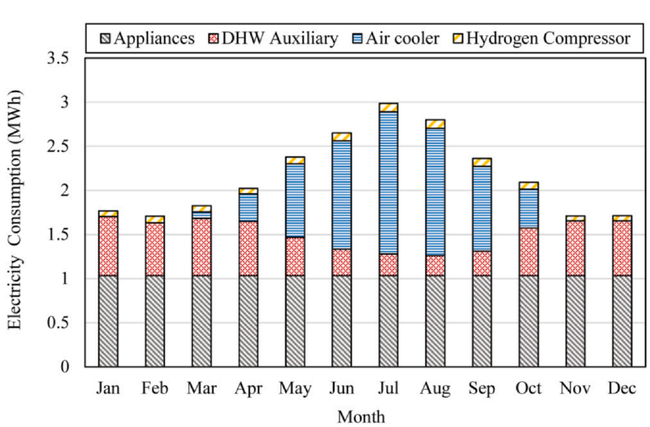

Fig. 7. Monthly electricity consumption profile.

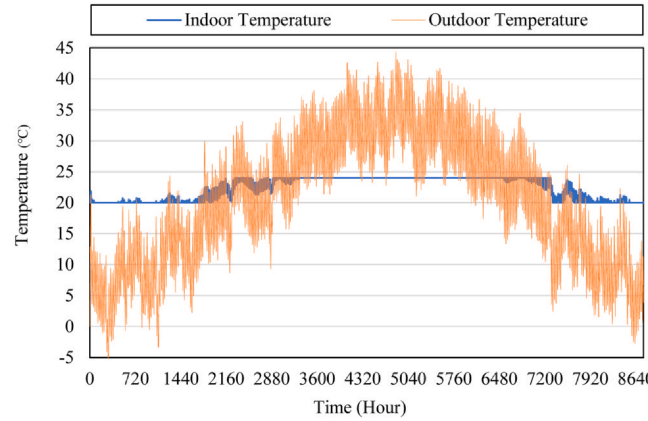

Fig. 8. Annual indoor and outdoor temperatures of the studied building.

Over the course of a year, the PVT system generates 56.75 MWh of electricity, from which 76.4% is surplus and is directed to the electrolyzer. The electrolyzer's total annual electricity consumption is calculated at 34.6 MWh, indicating that around 80% of the available surplus energy is effectively captured for storage. The PVT system contributes directly to 53.7% of the total annual electrical energy demand, with the remaining share supplied by the fuel cell, which produces 12.97 MWh of electricity annually. Fig. 9a presents the monthly distribution of power generation between these two sources. The PVT system delivers its highest share of electricity generation in April (62%), while its contribution reaches a minimum (45.7%) in December, linked to the lower solar radiation during that period.

The thermal output of the PVT system, which totals 795 kWh annually, accounts for 22.29% of the annual space heating demand, while the biomass furnace provides the remainder. The monthly contributions of both systems to space heating are shown in Fig. 9b. The highest thermal contribution from the PVT system is found in March, attributed to the high solar radiation and outdoor temperatures. The relatively low thermal contribution of the BIPVT system also highlights the importance of heat-transfer enhancement within its rear air channel. In this context, Yan et al. [54] showed that extended surfaces such as fins can increase the effective heat-transfer area and strengthen convective heat transfer, which is directly relevant to improving thermal extraction from air-based BIPVT systems.

Fig. 9c presents the monthly contributions of the fuel cell and the auxiliary element in meeting the DHW demand. The waste thermal output of the fuel cell supplies 52% of the annual DHW demand. As shown, its highest monthly contribution occurs in July and August, when the fuel cell operates more intensively to compensate for the power shortage not covered by the PVT system.

The monthly biomass consumption is shown in Fig. 10. A total of 1117 kg of sheep manure is used to compensate for the PVT's reduced output. The greatest consumption, 393 kg, takes place in January when the space heating demand is highest.

The contribution of the three main sources to the overall energy of the studied RHC is shown in Fig. 11. The PVT system meets half of the total demand, the generated hydrogen from the HES unit covers 40.3% of it, and the biomass accounts for the remaining 9.8%.

In addition to the electrical and thermal energy, the proposed energy system also generates oxygen gas as a by-product from the electrolysis process. Over the course of a year, a total of 3755 m^3 of oxygen are produced. The hourly production rate can be found in Fig. 12. The peak oxygen generation takes place on September 23, reaching 2087 L/h. This peak coincides with the highest monthly electrical energy consumption of the electrolyzer in September, equal to 3669 kWh. The operating status of the electrolyzer and the corresponding SOC profile of the hydrogen tank throughout the year are presented in Fig. 13.

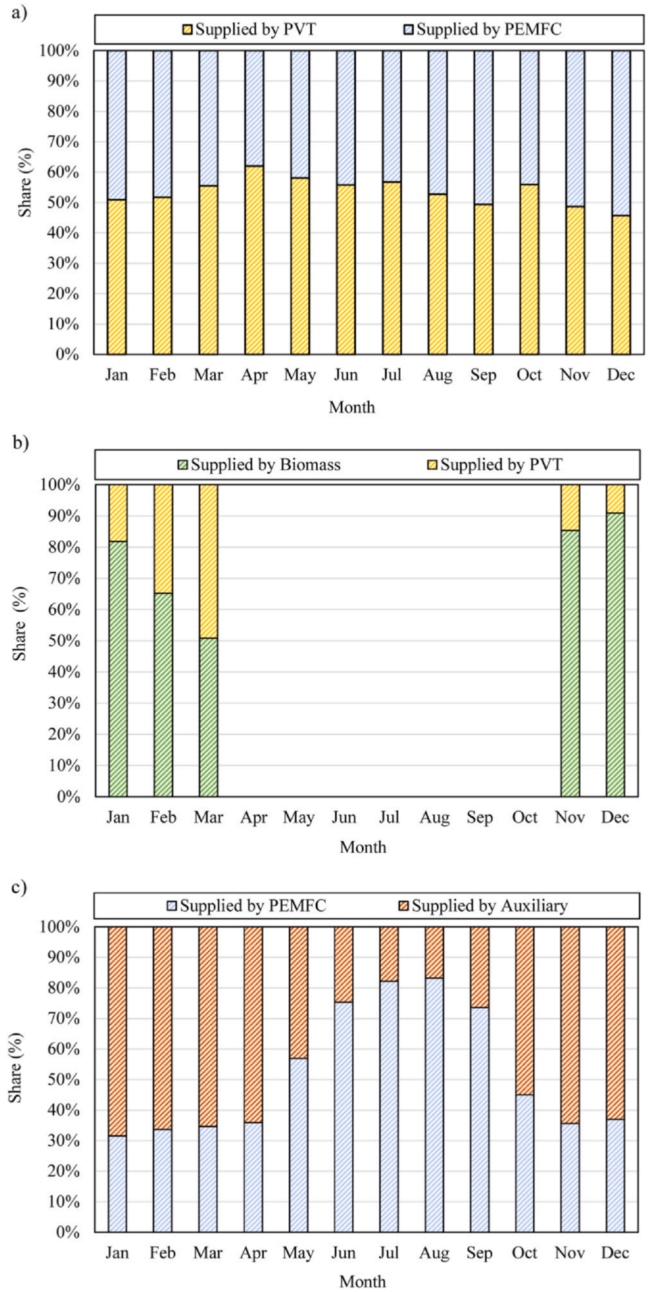

Fig. 9. The contribution of different sources to a) electricity, b) space heating, and c) DHW.

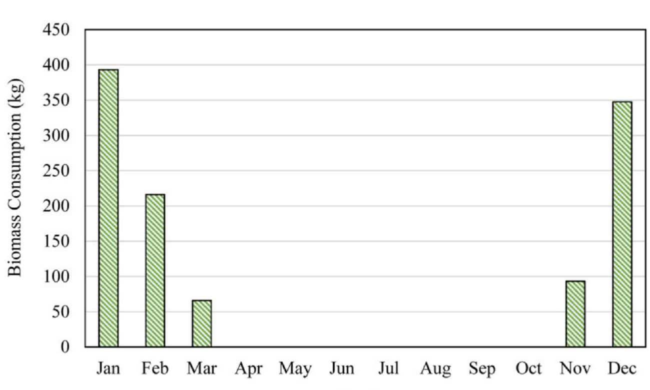

Fig. 10. Monthly biomass consumption profile.

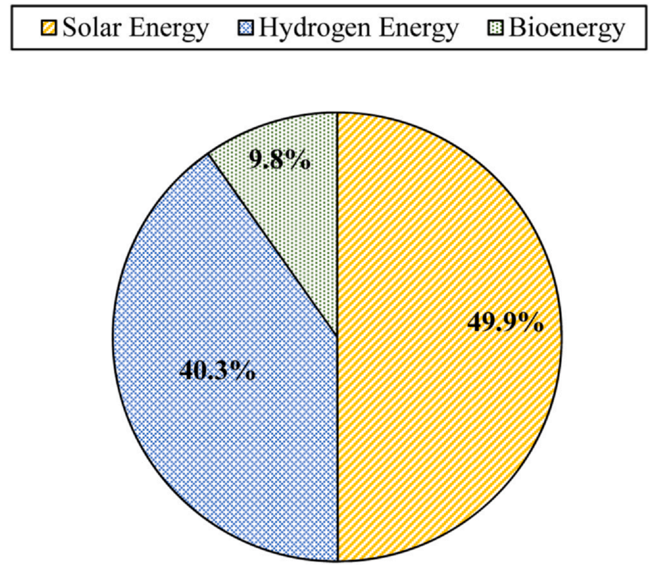

Fig. 11. The contribution of the different energy sources to the total annual energy demand.

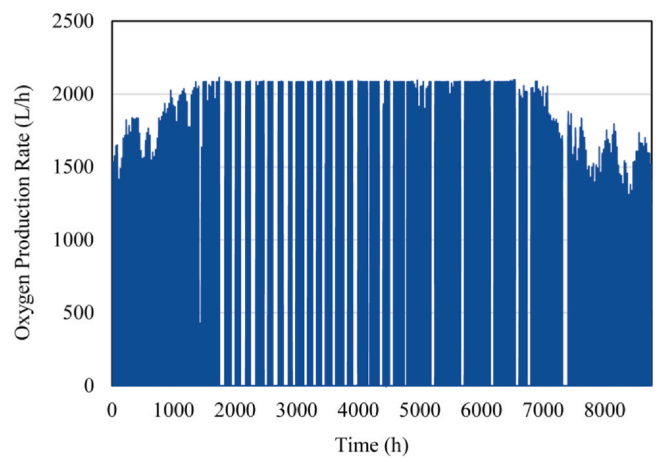

Fig. 12. Hourly oxygen production rate throughout the year.

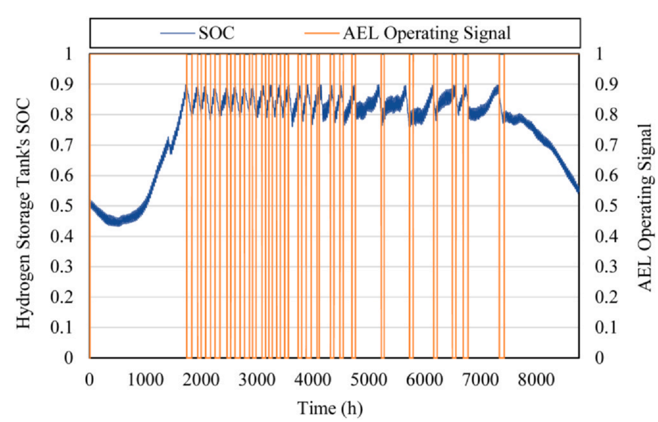

Fig. 13. Annual profile of the operation of the electrolyzer and the SOC of the hydrogen storage tank.

Assuming an average oxygen consumption of 7 L/min per patient and considering that the RHC is equipped with two oxygen-supported beds, the total oxygen requirement for continuous operation over a six-month period is estimated at 3679 m^3. This demand can be fully met by the oxygen produced with the electrolyzer. However, it should be noted that actual

Table 6
Benchmarking of selected technical performance indicators against comparable studies.

Study	System configuration	Load supply	PV/BIPVT electrical capacity factor (%)	Electrolyzer capacity factor (%)	H_2 production energy intensity (kWh/kg H_2)
Present study	BIPVT + AEL + H_2 tank + PEMFC + biomass furnace	Designed to meet full electrical and thermal load, achieving overall LOLP of 0.37%	24	20.8	51.8
[32]	PV + wind + PEM electrolyzer + H_2 tank + fuel cell	Designed to meet full electrical load	20.0	25.5	46.4
[60]	BIPV + electrolyzer + H_2 tank + fuel cell	Designed to meet electrical load with 2% capacity-shortage allowance	17.4–20.4	17–25	NR

oxygen requirements depend strongly on various clinical and operational factors and cannot be determined with absolute precision.

The TRNSYS Type 160a model calculates oxygen production stoichiometrically and does not predict the purity of the generated oxygen stream. In practical alkaline water electrolysis, oxygen purities of approximately 99.0–99.5 vol% without downstream purification have been reported [55]. Although some recent studies have considered electrolysis-derived oxygen suitable for medical applications without additional purification [56], oxygen purity can decrease under unfavourable operating conditions, particularly at low current densities and part-load operation, and is also influenced by electrolyte temperature, concentration, flow rate, and process management [57]. Accordingly, where the oxygen stream does not satisfy the applicable medical-gas specifications and standards, post-treatment methods such as catalytic hydrogen recombination and molecular-sieve drying can be employed to increase its purity to levels suitable for medical use [58].

3.1.1. Literature benchmarking

To further assess the consistency of the simulated system performance, selected technical indicators (BIPV/PV capacity factor, electrolyzer capacity factor, and hydrogen-production energy intensity) were benchmarked against two comparable off-grid renewable-hydrogen systems developed for rural healthcare facilities. These parameters were selected because they are normalized performance indicators that are relatively independent of system scale and can provide a meaningful basis for comparison across systems with different capacities and load levels. Table 6 shows that the obtained values are generally consistent with those reported in the literature. In particular, the calculated hydrogen production energy intensity falls within the commonly reported range of 40–60 kWh/kg H_2 [59].

3.2. Economic assessment

Considering a discount rate of 5%, the proposed system achieves a TLCC of $141,670 and delivers energy at an LCOE of $0.3/kWh with an LOLP

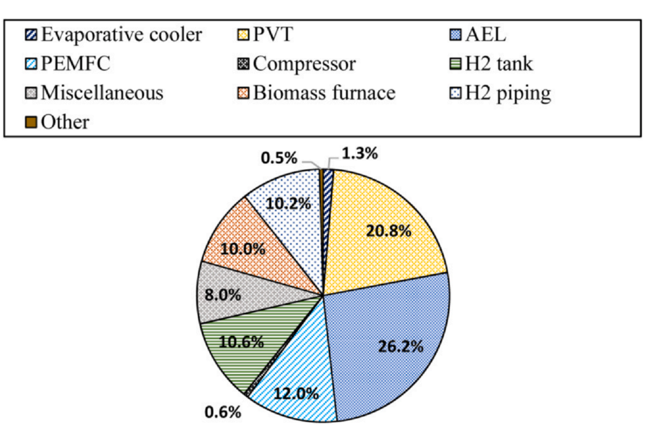

Fig. 14. The contribution of system components to the TLCC.

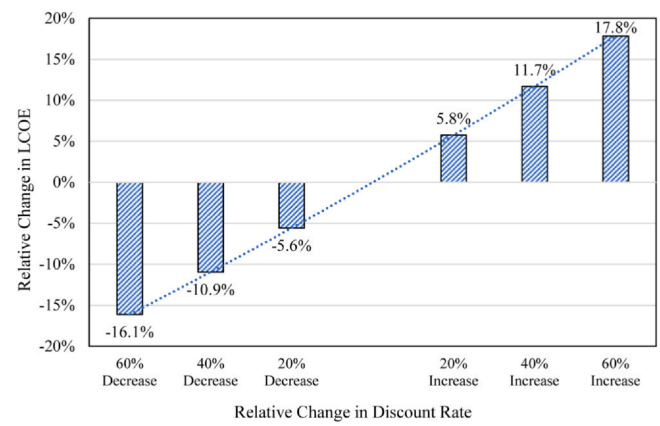

Fig. 15. Sensitivity of LCOE to variations in discount rate.

close to zero. These figures highlight the promising cost-effectiveness and robustness of the proposed design. Previous studies have reported LCOEs in the range of $0.21/kWh to $2.34/kWh, mostly supplying only electricity.

Fig. 14 illustrates the contribution of each system component to the TLCC, accounting for both capital and operational expenditure. The electrolyzer is the most significant cost driver, representing 26.2% of the TLCC, followed by the PVT system at 20.8%.

Considering the annual consumption of 1117 kg of sheep manure, approximately $112 per year is allocated to purchasing this biomass from local farmers. This expense represents a total of $2240 over 20 years of operation. Even though this expenditure is relatively modest, it can still encourage local communities to collect biowaste, providing a small but tangible source of supplementary income.

Since the economic performance of the system is sensitive to key financial and component-related assumptions, a sensitivity analysis was performed on the discount rate, AEL capital cost, and AEL lifespan. The AEL was selected for component-specific sensitivity analysis because it represents the largest individual contribution to the TLCC. As shown in Fig. 15 and Table 7, the LCOE increases consistently with the discount rate. When the discount rate is varied from 40% to 160% of its base value, the relative change in LCOE ranges from −16.1% to +17.8%. For the AEL capital cost, setting the CAPEX to 50% and 200% of its base value results in relative LCOE changes of −15.0% and +29.9%, respectively. Similarly, setting the AEL lifespan to 50% of its base value results in a +20.6% relative change in LCOE, while increasing it to 200% of the base value results in a −10.0% relative change.

3.3. Environmental assessment

The implementation of the proposed energy system results in the reduction of diesel consumption of approximately 10,000 litres per year. The annual CFOE of the proposed system is 12.25 times lower than that of a conventional diesel scenario, i.e., 0.0437 kg CO_2eq/kWh vs. 0.5354 kg CO_2eq/kWh. The annual life-cycle emissions for the proposed system are 1.6 tons CO_2eq, compared with 19.8 tons CO_2eq for the diesel baseline; 77% of

Table 7
Sensitivity of LCOE to variations in AEL capital cost and lifespan.

Scenario	LCOE ($/kWh)	Relative change in LCOE
Base case	0.2982	0%
AEL CAPEX = 50% of base value	0.2536	−15.0%
AEL CAPEX = 200% of base value	0.3873	+29.9%
AEL lifespan = 50% of base value	0.3595	+20.6%
AEL lifespan = 200% of base value	0.2684	−10.0%

baseline emissions arise from electricity for appliances and space cooling, with the remainder from space heating and DHW.

4. Conclusion

This study presented a hybrid polygeneration system integrating photovoltaic-thermal collectors, hydrogen energy storage, and a biomass furnace to supply energy to an off-grid rural clinic in Yazd, Iran. By simultaneously addressing electrical and thermal demands through a BIPVT-biomass-hydrogen configuration, this work extends existing research on rural healthcare electrification toward fully renewable and high-reliability polygeneration solutions suitable for arid climates. Year-long TRNSYS simulations showed that the proposed system reliably met the total annual energy demand of the clinic achieving a very low loss of load probability of 0.37%, maintaining indoor thermal comfort within 20–24°C throughout the year. Domestic hot water demand was partially met using fuel cell waste heat, and oxygen generated as a by-product of electrolysis could sustain two oxygen-supported beds for six months.

The system achieved a levelized cost of energy of $0.3/ kWh, which is competitive for remote healthcare facilities that lack grid access. Furthermore, the inclusion of modest incentives for local manure collection provided a small but meaningful community revenue stream. The total life-cycle cost of approximately $141,670 is justified by the system's ability to deliver continuous electricity, heating, cooling, and hot water without recurring diesel fuel costs.

By replacing diesel generators with a solar-biomass-hydrogen energy mix, the clinic's carbon footprint decreased significantly from 0.535 kg CO_2eq /kWh to 0.0437 kg CO_2eq /kWh, representing a 92% reduction in greenhouse gas emissions.

Overall, these results confirm that hybrid renewable polygeneration systems offer a resilient, affordable, and low-carbon solution for energy supply in remote healthcare facilities. Beyond improving energy access, such systems enhance medical capacity, stimulate local economies, and contribute to broader sustainability goals.

5. Limitations and Future Recommendations

While the proposed system demonstrates promising technical, economic, and environmental performance, several aspects remain outside the scope of the present analysis. The CFOE comparison relies on literature-based life-cycle emission factors rather than site-specific data for rural Iran. Consequently, emissions associated with transportation to remote locations, ageing, and efficiency degradation are not explicitly considered, and the actual environmental impact may therefore be higher than estimated. Future assessments should incorporate local and site-specific performance data.

The present study also considers HES as the storage option without a direct comparison with battery-only or hybrid battery-hydrogen configurations. Future work should evaluate these alternatives to determine the most suitable balance between storage efficiency, reliability, and cost for off-grid healthcare facilities. In addition, optimization of the system control strategy under varying climatic conditions and demand profiles could further improve system performance and assess the broader applicability of the proposed configuration.

Nomenclature

A	Area (m^2)
CF	Carbon footprint $\left(\frac{gCO_{2eq}}{kWh}\right)$
E	Energy (kWh)
m	Mass (kg)
P	Power (kW)
Q_{sh}	Space heating demand (kWh)
Subscripts	
AEL	Alkaline electrolyzer
Bio	Biomass
comp	Compressor
DB	Diesel boiler
DG	Diesel generator
PVT	Photovoltaic-thermal
PEMFC	PEM Fuel Cell
ref	Reference scenario

CRediT authorship contribution statement

Fontina Petrakopoulou: Writing – original draft, Supervision, Conceptualization. **Mohammad Mahdi Asghari:** Writing – original draft, Methodology. **Maryam Karami:** Writing – original draft, Supervision, Conceptualization. **Zahra Piryaei:** Writing – original draft, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

See Tables A1 and A2.

Table A1
The electrical equipment and their operating schedule in the RHC under study [32,61]

App.	N	P (W)	Time of the day (h)																							
			1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12
Blood Chemical Analyser	1	40										●		●		●		●								
CD4 Machine	1	200											●		●		●		●		●					
Centrifuge	1	240										●		●				●								
Computer	4	150	◐	◐	◐	◐	◐	◐	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	◐	◐	◐
Dental Chair	2	150									●	●	●	●	●		●		●	●						
Dental Compressor	1	1300									●							●								
Haematology Mixer	1	30									●		●		●		●		●							
Haematology Analyzer	1	230										●		●		●		●		●						
Lab Autoclave	1	1400								●										●						
Lightnings	32	16	●	●	●	●	◐	◐	◐	◐	◐	◐	◐	◐	◐	◐	◐	◐	●	●	●	●	●	●	●	●
Microscope	3	25							●	●	●	●	●	●	●	●	●	●								
Mobile Charger	5	20			◐			◐		◐		●		●	●		●	●		●	●					
Modems	4	10	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Oxygen Concentrator	2	300											●	●												
Refrigerator Blood Bank	1	80	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Refrigerator-Vaccine	1	50	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Transmitter	1	30										●				●					●					
TV Set	3	75								●	●	●	●	◐	◐	◐	●	●	●	●	●	●	●	●		
Ultrasonic Sealer	1	150											●		●		●			●						
Ultrasound Machine	1	800										●			●			●								
Vacuum Aspirator	1	40									●		●		●		●		●		●					
VHF Radio Receiver	1	2	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Note: ● Full use (100%); ◐ Half use (50%); blank cells indicate no use.

Table A2
Technical specification of the HRES components

Parameter	Value
BIPVT	
Total area (m^2)	304
Nominal power (kW)	27
Cover Conductivity ($kJ/h.\ m.\ K$)	5.04
Cover Thickness (cm)	0.635
Channel Height (cm)	5.08
Inlet Flowrate ($\frac{kg}{s}$)	2
AEL	
Electrode area (cm^2)	50
Number of cells in series	50
Number of stacks in parallel	2
Nominal Power (kW)	19
Maximum allowable current density ($\frac{mA}{cm^2}$)	300
Operating temperature (°C)	50
Operating Pressure (bar)	7
PEMFC	
Electrode area (cm^2)	232
PEM thickness (cm)	0.0118
Number of cells in series	60
Number of stacks in parallel	2
Nominal power (kW)	5.5
Hydrogen & Air inlet pressure (bar)	3.068
Hydrogen stoichiometric ratio	1.15
Oxygen stoichiometric ratio	2.5
Operating temperature (°C)	80
Maximum allowable cell current density ($\frac{mA}{cm^2}$)	700
Other	
Hydrogen tank volume (m^3)	3
DHW tank volume (m^3)	0.5
Evaporative cooler nominal power (kW)	2.8
Evaporative cooler inlet Flowrate ($\frac{kg}{s}$)	1
Evaporative cooler saturation efficiency	0.9
Biomass furnace nominal heating capacity (kW)	5
Biomass furnace inlet Flowrate ($\frac{kg}{s}$)	1
Biomass furnace heating input ratio (HIR)	1.3
Number of hydrogen compressor stages	3

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